Тв.

$$\xi \sim p(x) = \begin{cases} \frac{\theta-1}{x^{\theta}}, & x \geq 1 \\ 0, & x < 1 \end{cases}, \quad \theta > 1$$

a) ОМП:

$$L(\theta) = \prod_{i=1}^{n} p(x_i, \theta) = \frac{(\theta-1)^n}{\prod_{i=1}^{n} x_i^{\theta}}$$

$$\ln L(\theta) = n \ln(\theta - 1) - \sum_{i=1}^{n} \theta \ln x_i$$

$$(\ln L)' = \frac{n}{\theta-1} - \sum_{i=1}^{n} \ln x_i = 0 \Rightarrow \frac{n}{\theta-1} = \sum_{i=1}^{n} \ln x_i \Rightarrow$$

$$\Rightarrow \hat{\theta} = 1 + \frac{1}{\frac{1}{n}\sum_{i=1}^{n} \ln x_i}$$

$$(\ln L)'' = -\frac{n}{(\theta-1)^2} < 0 \quad \forall \theta > 1 \Rightarrow \max.$$

b) $\Theta = (1; +\infty)$ – открытое

$p(x, \theta)$ – непр. дифф. по θ на Θ.

$$\int_1^{+\infty} \frac{\partial p(x,\theta)}{\partial \theta} dx = \int_1^{+\infty} \frac{1 \cdot x^{\theta} - (\theta-1) x^{\theta} \cdot \ln x}{x^{2\theta}} dx = \int_1^{+\infty} \frac{\ln x - \theta \ln x + 1}{x^{\theta}} dx =$$

$$= \frac{x^{1-\theta}}{1-\theta}\Big|_1^{+\infty} + \int_1^{+\infty} \ln x \cdot \frac{1-\theta}{x^{\theta}} dx = \frac{-1}{1-\theta} + \int_1^{+\infty} \ln x \, d(x^{1-\theta}) =$$

$$= \frac{-1}{1-\theta} + \frac{1}{1-\theta} = 0$$

$$I(\theta) = \int_1^{+\infty} \frac{\theta-1}{x^{\theta}} \left(\frac{1}{\theta-1} - \ln x\right)^2 dx = \int_1^{+\infty} \frac{\theta-1}{x^{\theta}} \left(\frac{1}{(\theta-1)^2} - \frac{2\ln x}{\theta-1} + \ln^2 x\right) dx =$$

$$= \int_1^{+\infty} \left(\frac{1}{x^{\theta}(\theta-1)} - \frac{2\ln x}{x^{\theta}} + \frac{\ln^2 x \cdot (\theta-1)}{x^{\theta}}\right) dx =$$

$$= \frac{1}{\theta-1} \cdot \frac{1}{1-\theta} \cdot x^{1-\theta}\Big|_1^{+\infty} - 2\int_1^{+\infty} \ln x \, d\left(-\frac{1}{(\theta-1)x^{\theta-1}}\right) + \int_1^{+\infty} \ln^2 x \, d(-x^{1-\theta})$$

$$= \frac{1}{(\theta-1)^2} - 2\left(\ln x \cdot \frac{1}{(1-\theta)x^{\theta-1}}\Big|_1^{+\infty} + \int_1^{+\infty} \frac{1}{(\theta-1)x^{\theta-1}} \cdot \frac{1}{x}\,dx\right) +$$

$$+\left(-x^{1-\theta} \cdot \ln^2 x\right)\Big|_1^{+\infty} + 2\int_1^{+\infty} \frac{x^{1-\theta}\ln x}{x}\,dx =$$

$$= \frac{1}{(\theta-1)^2} - 2\int_1^{+\infty} \frac{x^{-\theta}}{\theta-1}\,dx + 2\int_1^{+\infty} x^{-\theta}\ln x\,dx =$$

$$= \frac{1}{(\theta-1)^2} - \frac{2}{\theta-1} \cdot \frac{1}{1-\theta} \cdot x^{1-\theta}\Big|_1^{+\infty} + 2 \cdot \frac{1}{(1-\theta)^2} =$$

$$= \frac{1}{(\theta-1)^2} - \frac{2}{(\theta-1)^2} + \frac{2}{(\theta-1)^2} = \frac{1}{(\theta-1)^2}$$ – непр. и >0 на Θ.

$\Rightarrow$ модель регулярна

$$\int_1^{+\infty} \frac{\partial^2 p(x,\theta)}{\partial\theta}\,dx = \int_1^{+\infty} \frac{-x^{\theta}\ln x - x^{\theta}\ln x(\ln x - \theta\ln x + 1)}{x^{2\theta}}\,dx =$$

$$= \int_1^{+\infty} \frac{-\ln x - \ln^2 x + \theta\ln^2 x - \ln x}{x^{\theta}}\,dx = \int_1^{+\infty} \frac{-2\ln x + \ln^2 x(\theta-1)}{x^{\theta}}\,dx =$$

$$= \frac{2}{(\theta-1)^2} - \frac{2}{(\theta-1)^2} = 0$$

$p(x,\theta)$ – 2-ды непр. дифф на Θ $\Big|$ $\Rightarrow$ модель сильно регулярна

<u>Медиана:</u>

$$F(x_{0,5}) = 0{,}5 \Rightarrow \int_1^{x_{0,5}} \frac{\theta-1}{t^{\theta}}\,dt = 1 - \frac{1}{x_{0,5}^{\theta-1}} = \frac{1}{2} \Rightarrow$$

$$\Rightarrow x_{0,5} = 2^{\frac{1}{\theta-1}} = g(\theta)$$

$$\nabla g(\hat\theta) = 2^{\frac{1}{\theta-1}} \cdot \ln 2 \cdot \frac{-1}{(\hat\theta-1)^2} \qquad I^{-1}(\theta) = (\theta-1)^2$$

$$\sigma^2(\theta) = 2^{\frac{1}{\theta-1}} \ln 2 \cdot \frac{-1}{(\theta-1)^2} \cdot (\theta-1)^2 \cdot 2^{\frac{1}{\theta-1}} \ln 2 \cdot \frac{-1}{(\theta-1)^2} =$$

$$= 2^{\frac{2}{\theta-1}} \ln^2 2 \cdot \frac{1}{(\theta-1)^2}$$

$$\sqrt{n}\,\frac{2^{\frac{1}{\theta-1}} - 2^{\frac{1}{\theta-1}}}{2^{\frac{1}{\theta-1}} \cdot \ln 2 \,\frac{1}{\hat{\theta}-1}} \rightsquigarrow N(0;1)$$

$$t_1 = u_{\frac{1-\beta}{2}} \qquad t_2 = u_{\frac{1+\beta}{2}}$$

$$t_1 < \sqrt{n}\,\frac{2^{\frac{1}{\theta-1}} - x_{0,5}}{2^{\frac{1}{\theta-1}} \ln 2 \,\frac{1}{\hat{\theta}-1}} < t_2$$

$$2^{\frac{1}{\theta-1}} - t_2 \cdot \frac{2^{\frac{1}{\theta-1}} \cdot \ln 2}{\sqrt{n}\,(\hat{\theta}-1)} < x_{0,5} < 2^{\frac{1}{\theta-1}} - t_1 \cdot \frac{2^{\frac{1}{\theta-1}} \cdot \ln 2}{\sqrt{n}\,(\hat{\theta}-1)}$$

c) $g(\theta) = \theta \Rightarrow \nabla g(\theta) = 1 \qquad \sigma^2(\theta) = 1(\theta-1)^2 \cdot 1$

$$\sqrt{n}\,\frac{\hat{\theta}-\theta}{(\hat{\theta}-1)} \rightsquigarrow N(0;1) \qquad \hat{\theta}-1 = \frac{1}{\frac{1}{n}\sum_{i=1}^{n} \ln x_i}$$

$$t_1 = u_{\frac{1-\beta}{2}} \qquad t_2 = u_{\frac{1+\beta}{2}}$$

$$t_1 < \sqrt{n}\,\frac{\hat{\theta}-\theta}{\hat{\theta}-1} < t_2$$

$$\hat{\theta} - t_2 \cdot \frac{\hat{\theta}-1}{\sqrt{n}} < \theta < \hat{\theta} - t_1 \cdot \frac{\hat{\theta}-1}{\sqrt{n}}$$

$$\hat{\theta} - \frac{t_2}{\frac{1}{\sqrt{n}}\sum_{i=1}^{n} \ln x_i} < \theta < \hat{\theta} - \frac{t_1}{\frac{1}{\sqrt{n}}\sum_{i=1}^{n} \ln x_i}$$

Продолжение в Тб. ірдпb